

Computer Organization And Assembly Language

```
00000000: 01001101 01011010 10010000 00000000 00000011 00000000 MZ....
00000006: 00000000 00000000 00000100 00000000 00000000 00000000 .....
0000000c: 11111111 11111111 00000000 00000000 10111000 00000000 .....
00000012: 00000000 00000000 00000000 00000000 00000000 00000000 .....
00000018: 01000000 00000000 00000000 00000000 00000000 00000000 @.....
0000001e: 00000000 00000000 00000000 00000000 00000000 00000000 .....
00000024: 00000000 00000000 00000000 00000000 00000000 00000000 .....
0000002a: 00000000 00000000 00000000 00000000 00000000 00000000 .....
00000030: 00000000 00000000 00000000 00000000 00000000 00000000 .....
00000036: 00000000 00000000 00000000 00000000 00000000 00000000 .....
0000003c: 10000000 00000000 00000000 00000000 00001110 00011111 .....
00000042: 10111010 00001110 00000000 10110100 00001001 11001101 .....
00000048: 00100001 10111000 00000001 01001100 11001101 00100001 !..L.!
0000004e: 01010100 01101000 01101001 01110011 00100000 01110000 This p
00000054: 01110010 01101111 01100111 01110010 01100001 01101101 rogram
0000005a: 00100000 01100011 01100001 01101110 01101110 01101111 canno
00000060: 01110100 00100000 01100010 01100101 00100000 01110010 t be r
00000066: 01110101 01101110 00100000 01101001 01101110 00100000 un in
0000006c: 01000100 01001111 01010011 00100000 01101101 01101111 DOS mo
00000072: 01100100 01100101 00101110 00001101 00001101 00001010 de....
00000078: 00100100 00000000 00000000 00000000 00000000 00000000 $. ....
0000007e: 00000000 00000000 01010000 01000101 00000000 00000000 ..PE..
```

Lab Manual 08

Objectives:

1. To learn how mul and and div instructions work
2. To Print values of registers using mul/div instructions
3. To understand and apply how can we use mul and div instruction in decimal output

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LAB TASKS

NOTE: You are required to prepare the ASM file of the following task. Use maximum number of procedures to do the tasks.

NOTE: Read instructions carefully.

- Take inputs in hexadecimal format and give output in decimal.
- Write separate procedures for input and output, despite the main procedure.
- Use AX/DX registers for arguments/returns to/from the procedures.
- Write your procedures in such a way that it can be used from anywhere and preserves the registers' previous data.
- Use properly push and pop to avoid losing values of registers.

Task 1:

Write an assembly program which:

- Allows the user to give input in seconds upto $2^{16}-1$.
- Converts the seconds to hours, minutes and seconds.
- Display the output in the following format:

Hours: hh

Minutes: mm

Seconds: ss

Task 2:

Write an assembly program which:

- Allows the user to give input in cents C; $0 \leq C \leq 99$.
- Converts the cents to half dollars, quarters, dimes, nickels and pennies.
(Note: 1 Dollar = 2 Half Dollars = 4 quarters = 10 dimes = 20 nickels = 100 pennies = 100 cents)
- Display the output in the following format:

Half Dollars: 0/1

Quarters: 0/1

Dimes: 0/1/2

Nickels: 0/1

Pennies: 0-4